

STANLEY COUNTY AP, NC, USA

WMO: 722148

Lat: 35.417N

Lon: 80.151W

Elev: 186

StdP: 99.12

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63824

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.5	(c) -5.7	(d) -17.8	(e) 0.8	(f) -2.1	(g) -14.8	(h) 1.1	(i) -1.0	(j) 8.8	(k) 11.9	(l) 7.8	(m) 10.1	(n) 0.9	(o) 350	(p) 0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	35.3	23.7	33.7	23.9	32.6	23.7	26.0	31.6	25.4	30.9	24.9	30.1	2.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.0	28.2	24.0	19.3	27.5	23.5	18.8	27.0	81.7	31.8	78.9	30.9	76.5	30.3	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.6	(b) 6.5	(c) 5.5	(e) -11.6	(f) 37.2	(g) 2.6	(h) 2.4	(i) -13.5	(j) 38.9	(k) -15.1	(l) 40.2	(m) -16.5	(n) 41.6	(o) -18.4	(p) 43.3		
(a) 7.6	(b) 6.5	(c) 5.5	(e) -12.1	(f) 27.3	(g) 2.5	(h) 1.2	(i) -13.9	(j) 28.1	(k) -15.4	(l) 28.8	(m) -16.9	(n) 29.4	(o) -18.7	(p) 30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.7	6.5	10.3	15.5	20.1	24.6	26.1	25.6	22.6	16.0	9.9	6.9
	DBStd	8.67	5.34	5.26	5.11	4.20	3.62	2.66	2.11	2.43	3.01	4.36	4.45	4.88
	HDD10.0	554	177	122	61	7	0	0	0	0	0	5	56	126
	HDD18.3	1849	422	332	253	104	26	1	0	0	4	96	255	357
	CDD10.0	2666	14	25	69	171	312	439	501	484	378	192	52	28
	CDD18.3	919	0	1	3	17	80	189	242	226	133	24	1	1
	CDH23.3	8610	0	9	58	248	757	1840	2304	2052	1101	226	13	3
	CDH26.7	3567	0	0	6	44	243	804	1066	922	431	49	0	0
Wind	WSAvg	1.8	1.9	2.2	2.4	2.4	1.9	1.6	1.4	1.4	1.8	1.7	1.7	1.7
Precipitation	PrecAvg	1221	99	95	122	83	104	116	128	115	107	87	79	87
	PrecMax	1626	256	209	293	198	254	297	408	319	293	410	189	206
	PrecMin	744	18	18	30	17	30	2	20	9	1	0	10	11
	PrecStd	193	53	46	60	43	52	65	86	66	74	74	44	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	24.8	27.4	29.7	33.1	36.3	36.7	37.8	35.0	30.7	25.0	22.8
		MCWB	15.4	17.4	16.6	19.4	22.4	23.3	24.4	22.7	21.5	23.2	17.2	18.6
	2%	DB	18.4	21.9	25.0	27.8	31.3	34.4	35.1	35.1	32.7	27.7	22.6	20.2
		MCWB	14.6	15.3	16.1	18.4	21.6	23.5	24.6	24.2	22.3	20.3	16.5	17.6
	5%	DB	16.3	18.7	22.7	26.3	29.3	32.8	33.2	32.8	31.4	25.9	20.9	17.9
		MCWB	13.0	13.9	14.8	17.1	21.4	23.5	24.5	24.4	22.5	19.0	16.2	15.7
	10%	DB	13.8	16.4	19.9	24.3	27.6	31.4	32.3	31.8	29.0	23.9	18.2	15.9
		MCWB	10.9	11.4	13.8	16.8	20.3	23.1	24.2	24.2	22.0	19.3	14.1	12.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.9	18.7	19.1	22.2	24.1	26.2	27.2	26.8	25.3	23.9	20.9	20.9
		MCDB	19.2	23.0	24.2	26.3	29.1	31.8	33.4	32.7	29.8	27.1	21.7	21.3
	2%	WB	15.8	17.1	17.8	20.6	23.0	25.3	26.1	25.9	24.6	23.0	18.9	18.3
		MCDB	17.7	19.8	22.6	24.7	27.8	31.2	31.9	31.3	29.1	25.5	21.3	19.4
	5%	WB	13.8	15.1	16.6	19.5	22.3	24.6	25.5	25.3	23.8	21.6	17.3	16.4
		MCDB	15.5	18.3	20.2	23.9	27.3	30.1	31.1	30.4	28.2	24.4	19.0	17.9
	10%	WB	11.3	12.5	15.2	18.4	21.6	24.0	25.0	24.8	23.0	20.0	15.2	13.5
		MCDB	13.1	15.1	19.0	22.8	26.1	29.1	30.3	29.5	26.6	22.8	18.1	15.0
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.6	13.4	13.8	12.6	11.9	11.5	11.1	11.6	12.5	13.4	11.6
		MCDBR	14.6	16.2	16.6	16.3	14.8	14.0	13.3	13.3	14.7	14.7	15.7	13.2
		MCWBR	10.9	10.3	8.9	8.2	6.3	5.1	4.7	5.2	5.9	7.7	10.2	9.6
	5% WB	MCDBR	12.0	14.0	13.8	13.0	12.0	12.1	11.6	11.4	11.4	11.0	11.4	10.8
		MCWBR	10.6	10.5	8.8	7.6	5.9	5.0	4.7	4.9	5.2	6.8	9.2	9.0
Clear-Sky Solar Irradiance	taub	0.311	0.323	0.358	0.399	0.455	0.484	0.535	0.513	0.425	0.382	0.327	0.309	
	taud	2.507	2.469	2.402	2.284	2.167	2.123	2.006	2.052	2.290	2.385	2.498	2.545	
	Ebn at Noon	894	922	911	886	837	809	765	774	831	841	866	875	
	Edh at Noon	82	95	111	132	150	157	175	164	123	103	82	75	
All-Sky Solar Radiation	RadAvg	2.61	3.27	4.34	5.43	6.03	6.44	6.03	5.49	4.63	3.79	2.88	2.25	
	RadStd	0.16	0.40	0.31	0.41	0.49	0.47	0.39	0.26	0.41	0.49	0.23	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (69 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	+116

Nomenclature: See separate page